

- 10 (a)** e.g. infinite input impedance / resistance
 zero output impedance / resistance
 infinite gain
 infinite bandwidth
 infinite slew rate
(any three, 1 each) B3 [3]
- (b) (i)** with switch open, V^- is less (positive) than V^+ M1
 output is positive A1
 with switch closed, V^- is more (positive) than V^+ so output is negative A1 [3]
(allow similar scheme if V^- more positive than V^+ treated first)
- (ii)** 1. diodes connected correctly between output and earth M1
 2. green identified correctly A1 [2]
(do not allow this mark if not argued in (i))

11 (a) (i) $I / I_0 = \exp(-1.5 \times 2.9)$ C1